Problem

In the tradiy education industry most platforms have unstructured content like random videos or signals. There's no structured ciculum.
So I built Market makers to provide a structured LMS platform where a verified contributors can create courses and learners can follow a guided ciculum with progress tracking and examination.

Tech Stack (Mern)

React - It's allows fast development and responsive UI with reusable components.

Backend - NodeJS and Express allow multiple request together and easy to scale RESTfull Api

Database - I used MongoDB, because the course structure is hierarchial.
Course → modules → modules contains lessons which fits well with Document Database
JSON type NoSQL

⇒ Architecture

- ⇒ It's a full stack modular Architecture
- ⇒ Frontend communicates with backend throught Rest API's
- ⇒ Backend's structured into routes, middleware, and database models which improves scalability and maintainability.
- ⇒ Middleware handles authentication, validation and role based access control before it reaches controller

⇒ Database Design

Db consists of several main modules including users, course, module, Lessons, Progress, Exam and Exam attempt.

Relationships

Each course → user id
module → course id
lessons → module id
Progress → user Id + lesson Id
Exam Attempt → user id + Exam Id

* Walkthrough (How Website Works)

-> User Registration

User has to Register with their basic details like email, password. Now here we have two Roles learner and Contributor, so if we choose learner you will get register and if contributor it will take extra information and set you account status to pending this must be approved my admin then only you will get registered.

-> User login

After registration when user login with their email and password, backend verifies that with the password with the password in DB and a JWT token's generated on successful login and stored in local storage these tokens are used when we try to fetch protected routes that time these tokens are verified.

It ensures that only authenticated users can access protected features like course creation, exams and progress tracking.

Contributor Approval workflow

New Contributor status is pending, Now Admin check the contributor details, verifies and then change the status to accepted or rejected. Once accepted free to create educational content. These shows the role based access control.

Course Creation & learning structure

Each accepted contributor can create courses containing multiple modules then contain ordered learning lessons.

This hierarchy ensures that students will learn structured circulum rather than random content.

Lessons Progress Tracking

When learner completes a lesson, the backend stores the progress. It's done using compound indexing in Progress db where we have lessonId and UsersId and it's kept unique so that these will not be duplicate progress.

In learned dashboard we can calculate completition percentage based on complete lessons

Exam system

I have created a Exam db where we have a question array in which question, option, correct answer index is there

I have created Examdb where we have array of questions Each elements in that array is an object consists of questions, option, correct Answer.

I have created Exam Attempt db where when learner gives the Exam selected Answer and question Index is stored

then is compared with Answer in Examdb

Score
= correct Answers / Total questions * 100

I have kept it if the user passes the exam he become contributor

Challenges

As it's a hierarchial structure we need to ensure proper ordering between courses, modules and lessons like no two modules can have same name in one course. Modules order is unique within a course. All these was implemented using unique compound indexing and order.

Technical Things I learned like

- RBAC
- Async handler
- Error handler
- Compound indexing
- Pagination also
- JWT authentication